

Siddharth Gummadapu

732-829-4713 | sg97@illinois.edu | www.linkedin.com/in/siddharth-gummadapu/ | siddharth224.github.io/

EDUCATION

University of Illinois Urbana-Champaign

Urbana, IL

Bachelor of Science in Mechanical Engineering, Bachelor of Science in Computer Engineering, GPA: 3.87/4.00 May 2027

EXPERIENCE

Rocket Lab

Aug 2026 – Nov 2026

Spacecraft Structural Analysis Intern

Long Beach, CA

Boeing

May 2026 – Aug 2026

Structures Engineering Intern

Everett, WA

- Mitigated a 777X build hazard by redesigning airframe components in CATIA, narrowing an assembly gap by 50%
- Integrated redesign into production through ASME Y14.5 drawings, rework instructions, and alignment pilot holes
- Authored a 777-8F structural manual detailing header beam components and materials to support factory repairs

BAE Systems

Jun 2025 – Aug 2025

Hardware Engineering Intern

Nashua, NH

- Cut cost of prototyping rigid-flex backplanes by \$5,000 per unit via 3D printing for enclosure fit verification
- Designed ESD-safe, 3D-printed connectors to replace metal fasteners for avionics component testing, validated using ANSYS modal and random vibration FEA and force-gauge testing, achieved a factor of safety of 3
- Performed tolerance stack-up and thermal interface analysis for 7 top-cooled SMT components on a PCB assembly, defined TIM selection and EMI gasket compression ranges accounting for tolerances in avionics packaging
- Analyzed material compatibility for 8 components, specifying corrosion-resistant finishes (ENIG, chemfilm, nickel)

PROJECTS

Propeller and Test Stand | CAD, CFD, Python, Modal Analysis, Structural Analysis

Apr 2026 – Aug 2026

- Developed a BEMT solver in Python, predicting 7.4 N of thrust at 21,000 RPM for a custom nylon propeller
- Engineered a load cell thrust test stand with waterjet-cut blocks and bolted joints to validate predicted thrust
- Validated the test assembly through prestressed static and modal FEA with bolt pretension, frictional contacts, and mesh convergence, maintaining a 32.7% frequency margin from the 1,050 Hz blade-pass excitation

Mini Robotic Car | CAD, FEA, Mechanism Design, Laser Cutting, 3D Printing

Jan 2026 – May 2026

- Built a robotic vehicle with a shifting drivetrain to optimize performance across speed, strength, and agility
- Engineered a frame with springs for impact absorption, validated with hand calculations and FEA with FS of 2.5
- Prototyped a fully 3D-printed servo-actuated drum brake, validated performance with analysis and testing

LEADERSHIP

EV Concept Car

Aug 2023 – Current

Drivetrain Team Lead

Urbana, IL

- Directing a 5-member team, optimizing power delivery systems for an electric car in the Shell Eco-marathon
- Designing an aluminum motor mount and analyzing threaded-joint failure modes to verify structural integrity
- Developing a 16:1 chain and sprocket drivetrain to align motor and wheel RPM, machining custom components

American Society of Mechanical Engineers

Aug 2023 – Current

Director of Product Development

Urbana, IL

- Leading a 40 member committee developing engineering solutions for clients in industry and academia
- Overseeing projects including assistive devices and robotic projects with Avea Robotics (Y Combinator-backed)
- Organizing and teaching CFD and FEA workshops to over 100 students, expanding technical training within UIUC

TECHNICAL SKILLS

Mechanical: Geometric Dimensioning and Tolerancing (GD&T), Design for Manufacturing and Assembly (DFMA), Finite Element Analysis (FEA), Computer Aided Design (CAD), Computational Fluid Dynamics (CFD), 3D Printing

Platforms: Siemens NX, Nastran, SolidWorks, Creo, CATIA, Ansys/Fluent, MATLAB/Simulink, Vivado, KiCAD

Computer Languages: Python, C/C++, SystemVerilog, Java, JavaScript, HTML/CSS, VBA